Billboard x Spotify: Streaming Performance vs Chart Success

Stakeholders: Marketing teams, A&R (Artist and Repertorie), data analysts within the music industry

Topic:

This case study will explore the relationship between streaming performance and its Billboard chart success.

Problem:

The music industry relies heavily on predicting which songs will become successful. However, it is not always clear whether streaming performance and audio characteristics are reliable indicators of chart success.
This analysis aims to determine which factors are most associated with a song becoming a Billboard hit.

Business task:

Identify whether streaming metrics and audio features can help explain or predict a song's success on the Billboard charts.

Metrics:

The following metrics will be used to evaluate success:

- Spotify popularity score
- Billboard chart position
- Presence on Billboard charts (hit vs non-hit)
- Audio features (danceability, energy, acousticness, etc.)

Audience:

Music industry professionals are involved in decision-making related to artist promotion and release strategies.

Insights:

The insights from this analysis can help stakeholders:

- Identify characteristics of successful songs
- Improve marketing and promotion strategies.

- Better allocate resources to songs with higher potential for success.

Prepare

Data Sources

The data used in this analysis comes from two publicly available datasets hosted on Kaggle:

- **Spotify Songs Analytics Dataset:** Contains information about songs available on Spotify, including audio features such as danceability, energy, acousticness, and other attributes related to musical characteristics.
- **Billboard Historical Charts Dataset:** Includes weekly chart rankings, song titles, artist names, and dates of appearance on the Billboard charts, distributed across multiple CSV (Comma-Separated Values) files.

Data Location and Organization

Both datasets were downloaded from Kaggle and stored locally. Copies of the original files were created to ensure that the raw data remains unchanged during the data cleaning and analysis process.

The Spotify dataset is organized at the song level, while the Billboard dataset is structured as time-series data, with each row representing a song's position on the chart for a specific week.

Data Credibility (ROCCC)

- **Reliable:** The datasets are commonly used for educational and analytical purposes, although they are not official API exports.
- **Original:** The data is aggregated from Spotify and Billboard sources and shared via Kaggle.
- **Comprehensive:** Both datasets include a large number of records and relevant variables for analysis.
- **Current:** The Spotify dataset may not be fully up-to-date, which could limit real-time insights. The Billboard dataset is regularly updated; however, this analysis uses a static version at the time of project preparation.
- **Cited:** While the datasets are publicly available, the original data collection methods for Spotify are not fully documented. The Billboard data is assumed to be collected from its officially published charts.

Data Integrity Verification

To ensure data quality, the following steps will be performed:

- Checking for missing or null values
- Identifying and removing duplicate records
- Standardizing text fields such as song titles and artist names
- Reviewing data formats (e.g., dates and numerical values) for consistency

Data Relevance

The datasets are directly aligned with the business task:

- The Spotify dataset provides streaming-related metrics and audio features.
- The Billboard dataset provides official chart performance data.

Together, they enable analysis of the relationship between streaming performance and chart success.

Data Limitations and Potential Issues

- Data may not be fully up-to-date
- Inconsistencies in text fields (e.g., song titles, artist names)
- Formatting differences (e.g., date structures)
- Potential bias toward popular or widely distributed music

Data Accessibility, Privacy, and Licensing

- The Spotify dataset is licensed under **CC0 (Public Domain)** and is free to use.
- The Billboard dataset is licensed under **CC BY-NC-SA 4.0** and is used for non-commercial, educational purposes with proper attribution.
- Both datasets are publicly available on Kaggle.
- No personal or sensitive user data is included.
- The data is used in compliance with Kaggle's terms of use.

Process

For the Spotify dataset, Google Sheets and BigQuery were used for data cleaning and preprocessing.
For the Billboard dataset, BigQuery was used exclusively due to its larger size and

complexity. The Billboard Radio, streaming songs, and Digital songs CSV files were excluded, as this analysis focuses specifically on popular charts.

The Spotify dataset initially contained 955 rows and 24 columns, including variables such as track name, artist, number of contributing artists, release date components (year, month, day), playlist presence across Spotify, Apple Music, and Deezer, chart rankings across multiple platforms (Spotify, Apple Music, Deezer, Shazam), total streams, BPM, key, musical mode, and several audio features (danceability, valence, energy, acousticness, instrumentalness, liveness, and speechiness).

During the cleaning process:

- Blank spaces were identified and removed in cells B214 and B274.
- No duplicate records were found.
- No null values were detected.
- Row 576 was removed due to data inconsistencies that could not be reliably corrected.
- Row 376 was removed because it did not contain a valid track name, only unreadable symbols.
- A new column, **“release_date”**, was created by combining release day, month, and year to standardize date formatting across datasets. The original three columns were subsequently removed for clarity.
- Audio feature columns (danceability, valence, energy, acousticness, instrumentalness, liveness, and speechiness) were converted to percentage format for consistency.
- Missing values in the **Shazam Charts** column were replaced with 0.
- Missing values in the **key** column were replaced with “unknown”.

After these steps, SQL was used in BigQuery to finalize the cleaning process. A new column, **“clean_track_name”**, was created to standardize track names by removing unnecessary information such as featured artists:

```
CREATE OR REPLACE TABLE
capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned_final AS

SELECT

  *,

  TRIM(REGEXP_REPLACE(track_name, r'\(\s*(?i:feat|ft|with|featuring)\b[^)]*\)',
'')) AS clean_track_name

FROM capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned
```

The Billboard dataset comprises multiple CSV files, each representing a different chart category (For this analysis, we're gonna use only Billboard Hot 100) The dataset include:

- Billboard Hot 100 (350.79K rows)

All datasets share the same structure, containing identical columns: chart date, song title, artist, rank, last week's rank, peak position, weeks on chart, and image URL.

As part of the preprocessing phase, the column originally named **"date"** was standardized to **"chart_date"** across all tables to avoid conflicts with SQL reserved keywords and improve clarity.

A consistent data-cleaning process was applied to each dataset individually because of its identical structure. This process included:

- Duplicate detection
- Null value validation
- Blank space checks
- Data type and format validation
- Date validation
- Detection of unusual or inconsistent characters in text fields

After performing these checks, no null values were found in the dataset

All records were verified to be consistent with their expected formats, including dates and numerical fields. No duplicate entries were identified, and no issues related to formatting or unintended blank spaces were detected.

Overall, the Billboard datasets were determined to be clean, consistent, and suitable for further analysis.

Analysis

Spotify Popularity Score Construction

To quantify song popularity within the Spotify dataset, a composite score was created using three key metrics:

- Total streams
- Playlist presence across platforms
- Chart presence across platforms

Streams were given a higher weight (0.6) as they directly represent listener engagement, while playlist and chart presence (0.2 each) were treated as supporting indicators of exposure.

The following query was used to construct the score:

```
CREATE OR REPLACE TABLE
capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned_analisis_v1 AS
WITH base AS (
 SELECT
   track_name,
   artist_s__name,
   streams,
   release_date,
   (in_spotify_playlists + in_apple_playlists + in_deezer_playlists) AS
Total_playlist,
   (in_spotify_charts + in_apple_charts + in_deezer_charts + in_shazam_charts) AS
Total_charts,
   (
     LOG(streams + 1) * 0.6 +
     LOG(in_spotify_playlists + in_apple_playlists + in_deezer_playlists + 1) * 0.2
+
     LOG(in_spotify_charts + in_apple_charts + in_deezer_charts + in_shazam_charts
+ 1) * 0.2
   ) AS raw_score
 FROM capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned
)
SELECT
 *,
 100 * (
   (raw_score - MIN(raw_score) OVER()) /
   (MAX(raw_score) OVER() - MIN(raw_score) OVER())
 ) AS score_100
FROM base;
```

The `score_100` metric rescales the raw score into a range from 0 to 100, improving interpretability.

Initial exploration suggested that high streaming numbers do not necessarily correspond to a higher presence in charts or playlists, indicating that a single factor does not drive popularity.

Billboard Performance Metrics

To analyze chart performance, two definitions of success were established:

- **Peak performance** → highest rank achieved (`min_rank`)
- **Longevity** → total weeks on chart (`max_weeks_in_charts`)

The analysis was limited to 2020–2023 to align with the Spotify dataset.

```
CREATE OR REPLACE TABLE
`capstone-spotify-x-billboard.Billboard_Dataset.Bilboard_hot_100_analysis`
AS
SELECT
 Song,
 Artist,
 MIN(chart_date) AS first_chart_date,
 MAX(chart_date) AS last_chart_date,
 MIN(Rank) AS min_rank,
 MAX(SAFE_CAST(`Weeks in Charts` AS INT64)) AS max_weeks_in_charts
FROM `capstone-spotify-x-billboard.Billboard_Dataset.Bilboard_hot_100`
WHERE chart_date BETWEEN '2020-01-01' AND '2023-12-31'
GROUP BY 1, 2;
```

Data Normalization

To ensure accurate joins, song and artist names were standardized:

```
CREATE OR REPLACE TABLE
`capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned_analisis_v2` AS
SELECT
 *,
 INITCAP(TRIM(LOWER(REGEXP_REPLACE(track_name, r'\(.*?\)', '')))) AS Song,
 INITCAP(TRIM(LOWER(REGEXP_REPLACE(artist_s__name, r'\(.*?\)', '')))) AS Artist
FROM capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned_analisis_v1;
```

Spotify vs Chart Performance

A join between both datasets enabled direct comparison between Spotify Score and Weeks in Charts, and another comparison between Spotify Score and Rank

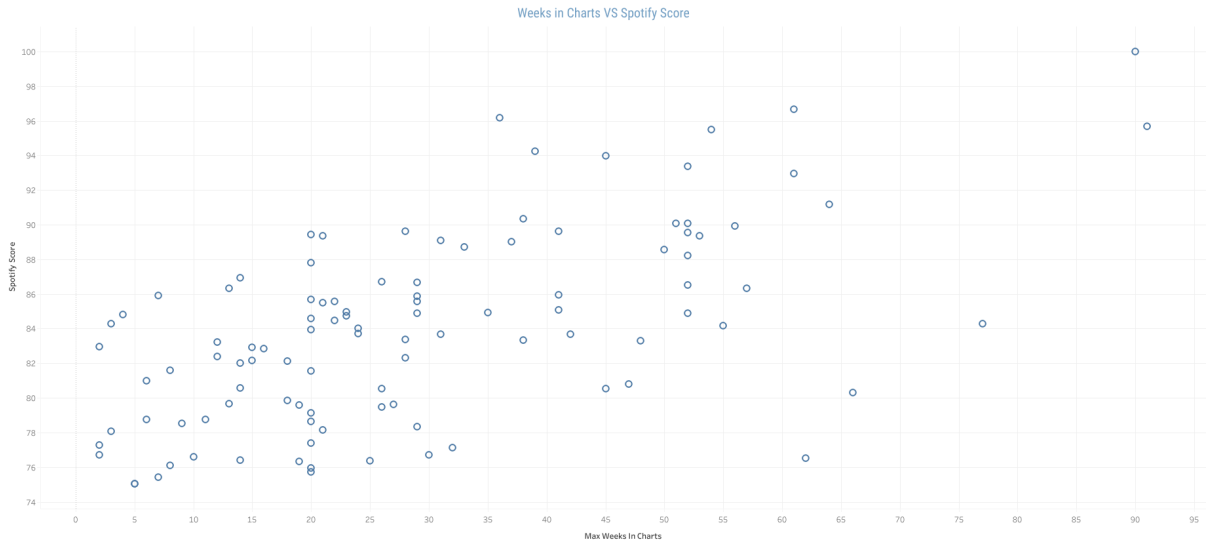
```
SELECT
 b.Artist,
 b.Song,
 b.min_rank,
 b.max_weeks_in_charts,
```

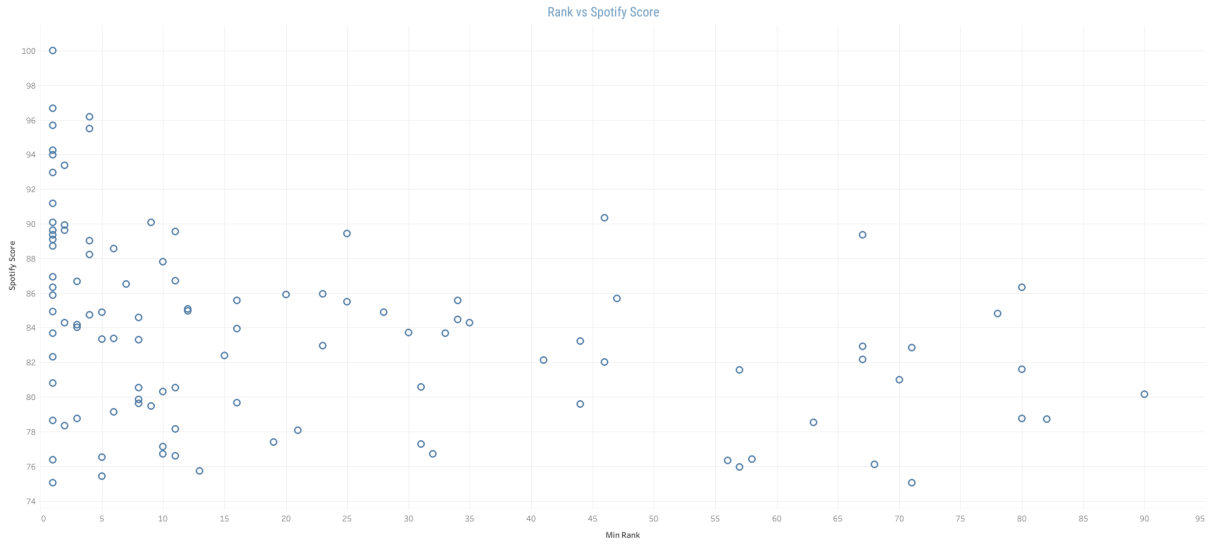
```
 b.first_chart_date,
 b.last_chart_date,
 s.score_100 AS spotify_score
FROM `capstone-spotify-x-billboard.Billboard_Dataset.Bilboard_hot_100_analysis` AS
b
INNER JOIN
`capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned_analisis_v2` AS s
ON b.Song = s.Song
AND b.Artist = s.Artist
ORDER BY b.min_rank ASC;
```

To quantify relationships:

Fila	corr_rank_vs_spotify	corr_weeks_vs_spotify
1	-0.34319281808116953	0.65739502258225357

```
SELECT
 CORR(s.score_100, b.min_rank) AS corr_rank_vs_spotify,
 CORR(s.score_100, b.max_weeks_in_charts) AS corr_weeks_vs_spotify
FROM `capstone-spotify-x-billboard.Billboard_Dataset.Bilboard_hot_100_analysis` b
INNER JOIN
`capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned_analisis_v2` s
ON b.Song = s.Song
AND b.Artist = s.Artist;
```





Key Findings

- Weak negative correlation between rank and Spotify score
- Moderate positive correlation between weeks on the chart and Spotify score

This suggests that **longevity is a better indicator of real popularity than peak position**.

Hit Classification

Songs were categorized based on performance to define what kind of success are in the Billboard Dataset:

- **Instant Hit** → high peak, short duration
- **Durable Hit** → long chart presence
- **Other**

```
SELECT
 CASE
   WHEN b.min_rank <= 10 AND b.max_weeks_in_charts <= 20 THEN 'Instant Hit'
   WHEN b.max_weeks_in_charts > 20 THEN 'Durable Hit'
   ELSE 'Other'
 END AS hit_category,
 AVG(s.score_100) AS avg_spotify_score,
 COUNT(*) AS total_songs
FROM capstone-spotify-x-billboard.Billboard_Dataset.Bilboard_hot_100_analysis b
JOIN capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned_analisis_v2 s
ON b.Song = s.Song
AND b.Artist = s.Artist
GROUP BY hit_category;
```

Fila	hit_category	avg_spotify_score	total_songs
1	Other	69.74120170865...	117
2	Instant Hit	74.49130128734...	24
3	Durable Hit	83.68776777633...	75

Insight

Durable hits consistently show higher Spotify scores, reinforcing the idea that sustained chart presence aligns with listener behavior.

Audio Features Analysis

To understand *how successful songs sound*, audio features were incorporated into a new table for a better and more separated analysis:

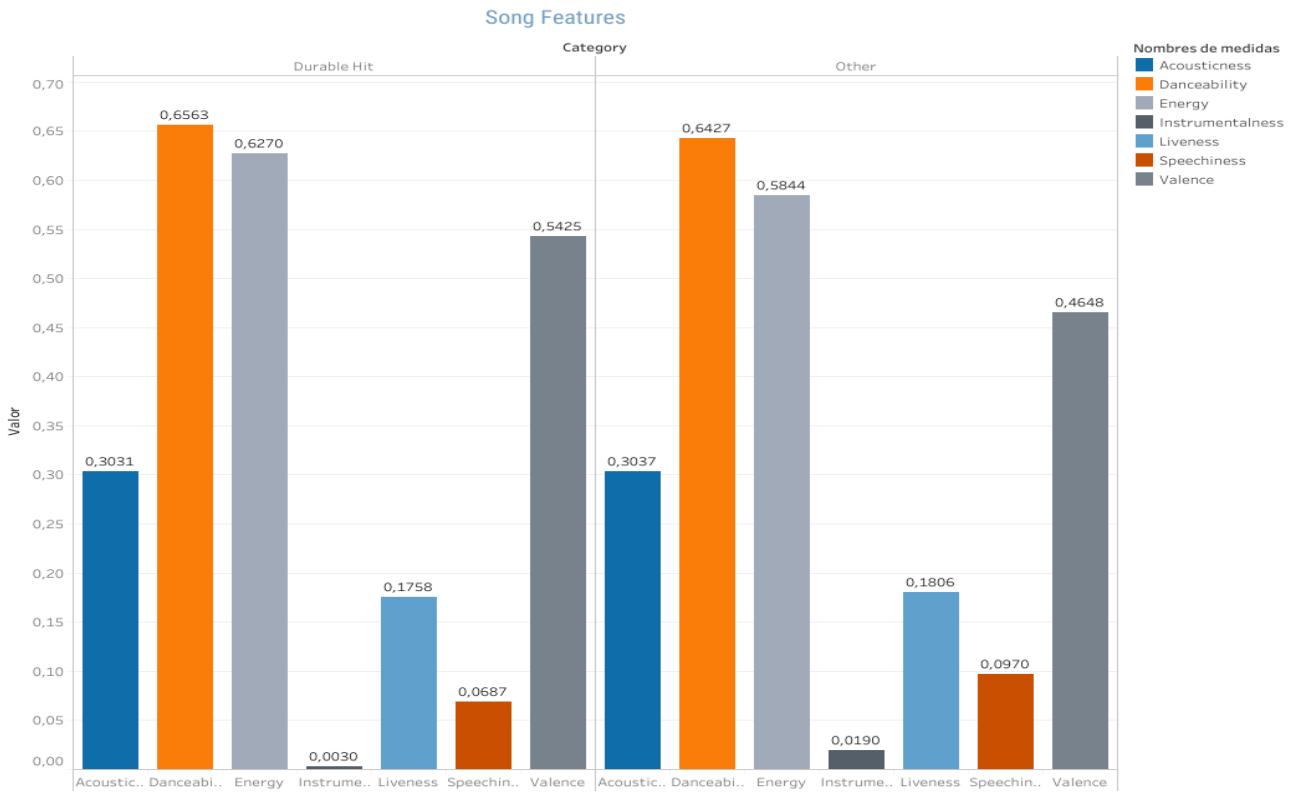
```
CREATE OR REPLACE TABLE
`capstone-spotify-x-billboard.Billboard_Dataset.Characteristics_hits`
AS
SELECT
 b.Song,
 b.Artist,
 MIN(b.min_rank) AS min_rank,
 MAX(b.max_weeks_in_charts) AS max_weeks_in_charts,
 AVG(s.`danceability_%`) AS danceability,
 AVG(s.`energy_%`) AS energy,
 AVG(s.`valence_%`) AS valence,
 AVG(s.`acousticness_%`) AS acousticness,
 AVG(s.`instrumentalness_%`) AS instrumentalness,
 AVG(s.`liveness_%`) AS liveness,
 AVG(s.`speechiness_%`) AS speechiness
FROM `capstone-spotify-x-billboard.Billboard_Dataset.Bilboard_hot_100_analysis` b
JOIN `capstone-spotify-x-billboard.Billboard_Dataset.Spotify_cleaned` s
ON b.Song = s.track_name
AND b.Artist = s.artist_s__name
GROUP BY b.Song, b.Artist;
```

Comparison by category:

```
SELECT
 CASE
   WHEN max_weeks_in_charts > 20 THEN 'Durable Hit'
   ELSE 'Other'
 END AS category,
 AVG(danceability) AS Danceability,
 AVG(energy) AS Energy,
 AVG(valence) AS Valence,
 AVG(acousticness) AS Acousticness,
 AVG(instrumentalness) AS Instrumentalness,
 AVG(liveness) AS Liveness,
 AVG(speechiness) AS Speechiness
FROM capstone-spotify-x-billboard.Billboard_Dataset.Characteristics_hits
GROUP BY category;
```

Fila	category	Danceability	Energy	Valence
1	Other	0.642699999999...	0.584366666666...	0.464800000000...
2	Durable Hit	0.656268656716...	0.627014925373...	0.542537313432...

Acousticness	Instrumentalness	Liveness	Speechness
0.303699999999...	0.019000000000...	0.180566666666...	0.097033333333...
0.303134328358...	0.002985074626...	0.175820895522...	0.068656716417...



Final Insight

Durable hits tend to be:

- More danceable
- More energetic
- More positive in tone
- Less acoustic and less instrumental

However, these differences are moderate.

Audio features contribute to success, but they do not fully determine it.

Conclusion

This analysis shows that:

- Peak chart position does not strongly reflect overall popularity
- Longevity is a more reliable indicator of sustained listener engagement
- Audio characteristics play a role, but are not decisive on their own

In short:

> **Success in music is not defined by a single metric, but by a combination of performance, consistency, and listener behavior.**